

**PAPER
C**



2020 SCIENCE

TIME ALLOWED: 55 MINUTES

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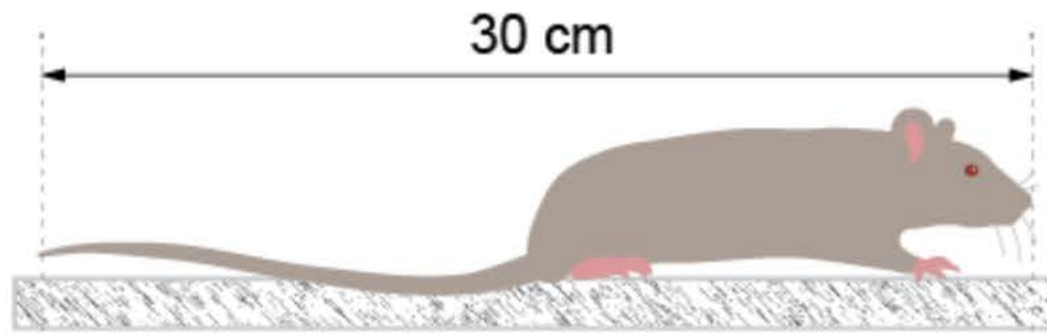
Solar panels convert energy from sunlight into electricity.



Which of the following would affect the amount of energy captured by the solar panels?

- ☐ how hot the day is
- ☐ how strong the roof is
- ☐ how much energy is needed
- ☐ how sunny it is during the day

This is a drawing of a rat.



About how many centimetres long is the rat's tail?

☐

5

☐

10

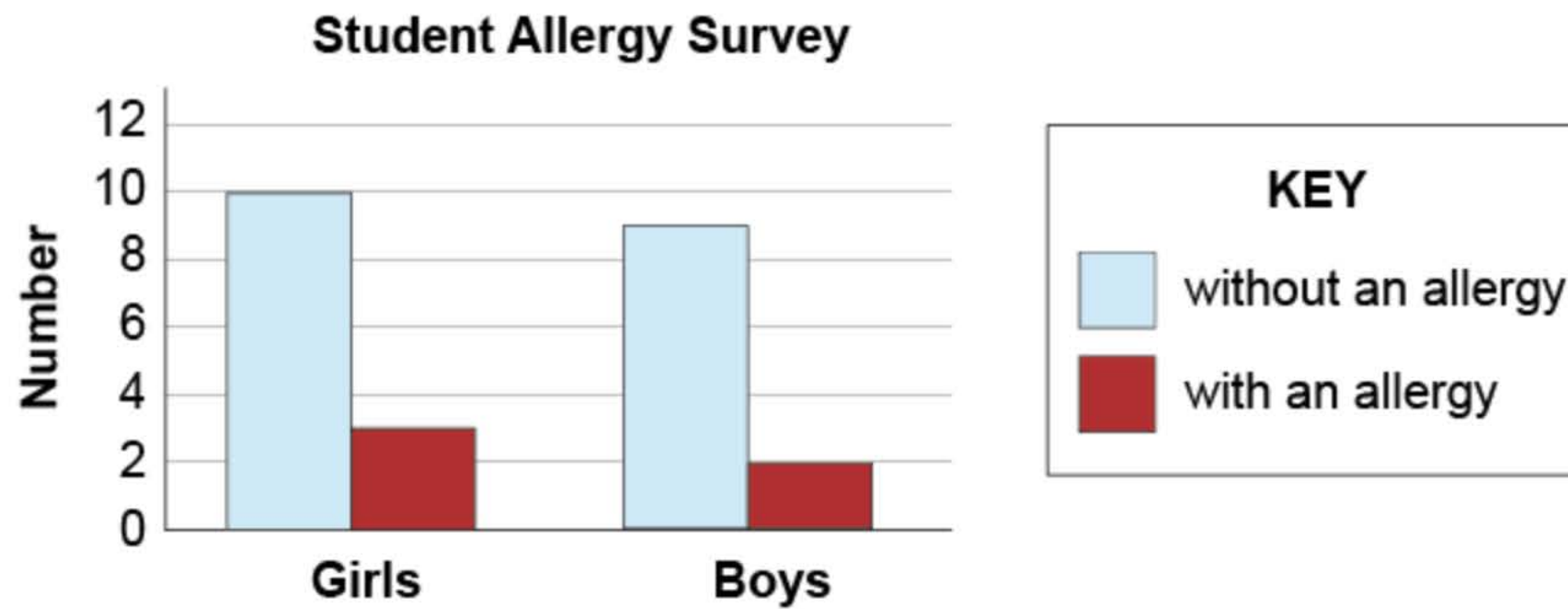
☐

15

☐

20

The graph shows the results of a student allergy survey in one class.



Which of the following groups has the greatest number of students in it?

☐

students with an allergy

☐

girls with an allergy

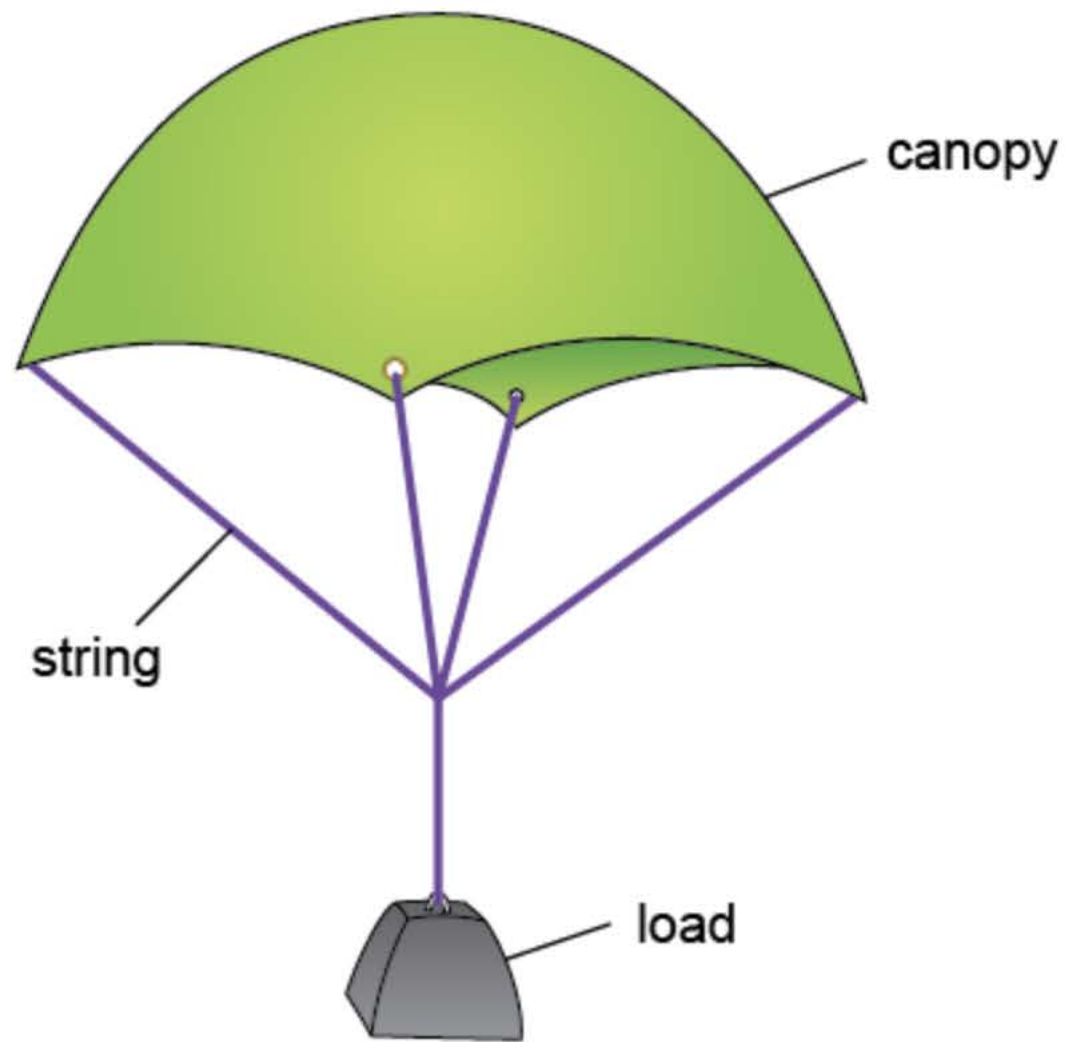
☐

boys without an allergy

☐

girls without an allergy

A simple parachute can be made of a canopy and some strings. Four students each made a parachute using different materials.



The students wanted to test which parachute would stay in the air the longest.

They took their parachutes to the second floor. They held their parachutes so that the load of each one was at the same height. Then they dropped their parachutes at the same time.

To make the test fair

☐

the canopies should be made from the same material.

☐

the strings should be the same length.

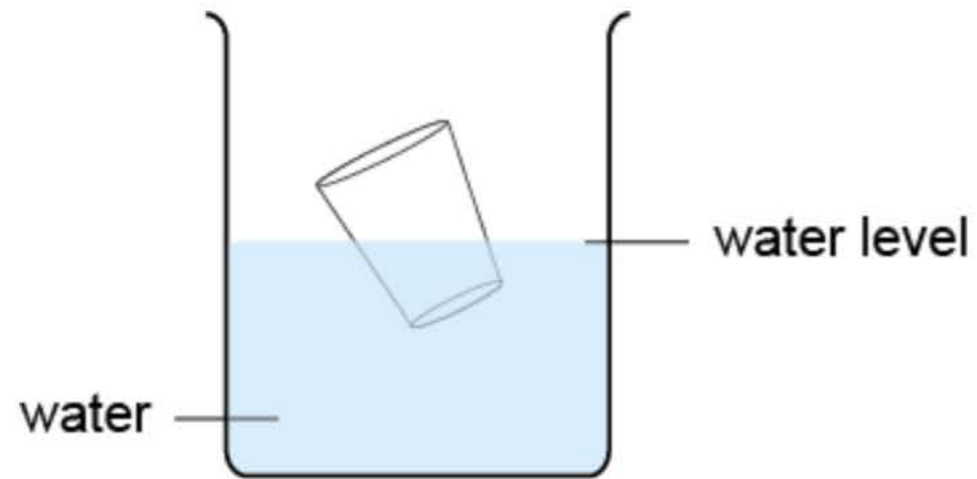
☐

the loads should be the same.

☐

the parachutes should weigh the same.

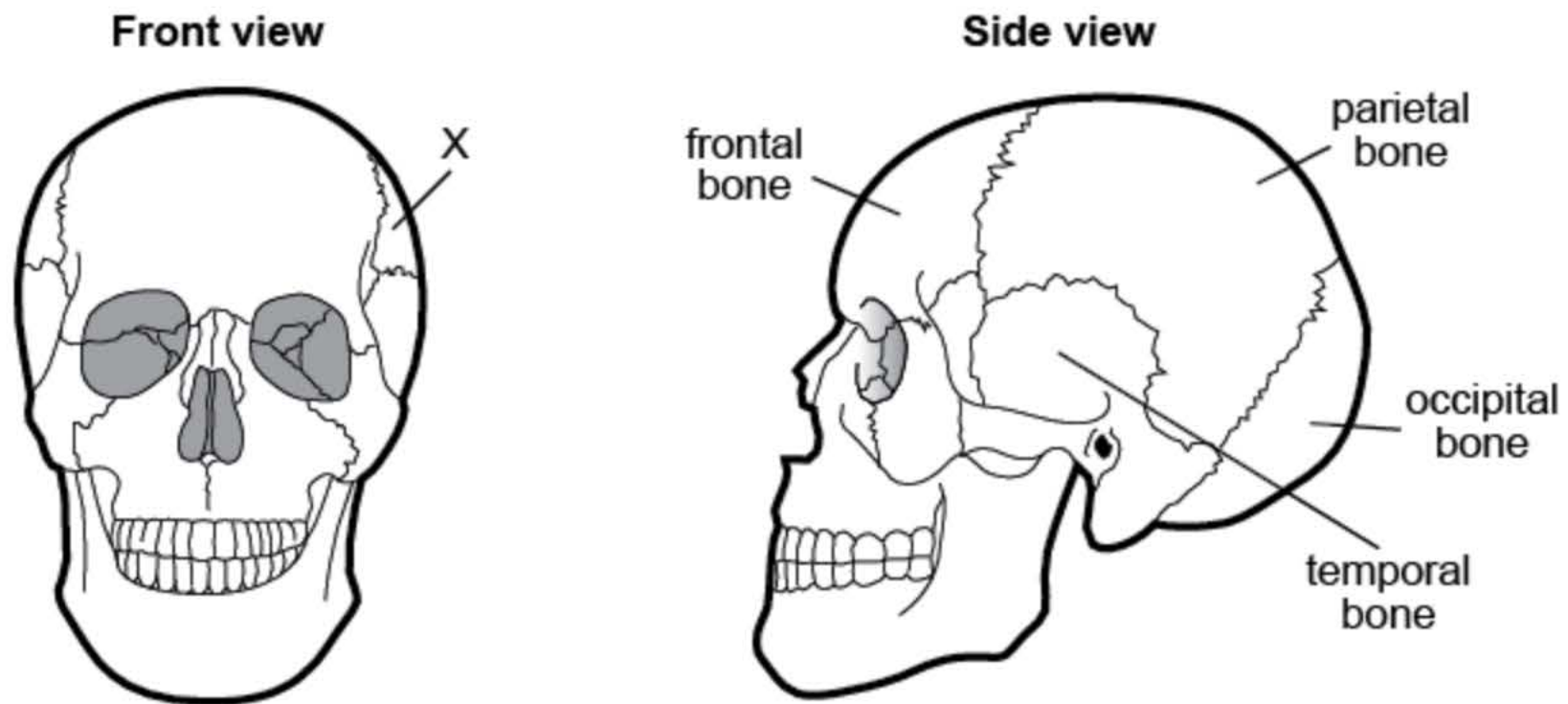
Lisa placed an empty plastic cup in a container of water.
The diagram shows the plastic cup floating in the water.



What will happen if Lisa places a marble into the plastic cup?

- ☐ The water level will move downwards.
- ☐ The water level will move upwards.
- ☐ The water level will not change.
- ☐ The amount of water in the container will increase.

The diagrams show the front and side views of a human skull.



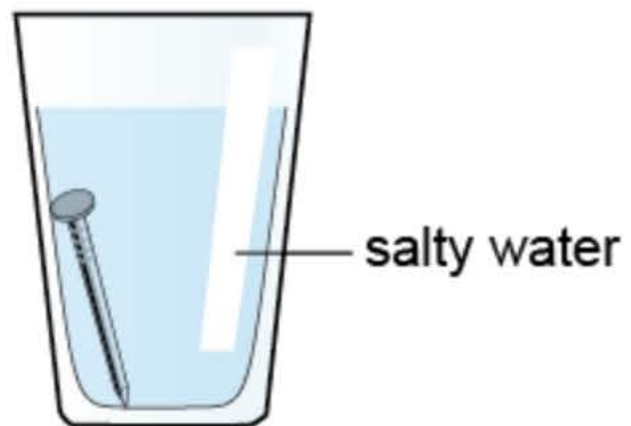
Which bone is X in the diagram of the front view of the human skull?

- | | |
|--|---|
| ☐ frontal bone | ☐ occipital bone |
| ☐ parietal bone | ☐ temporal bone |

Reddish brown flakes form on the surface of an iron nail when it rusts.



Freda put a clean iron nail in a glass of salty water.



Three days later the iron nail was rusty.

What can Freda learn from her investigation?

☐

Iron nails rust in salty water.

☐

Air stops iron nails rusting.

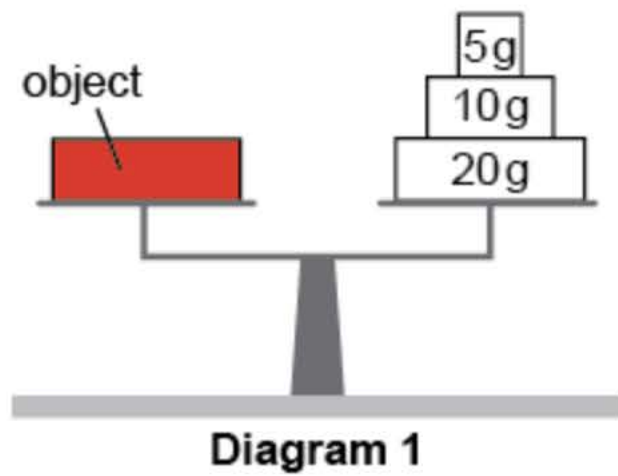
☐

Iron nails rust in warm water.

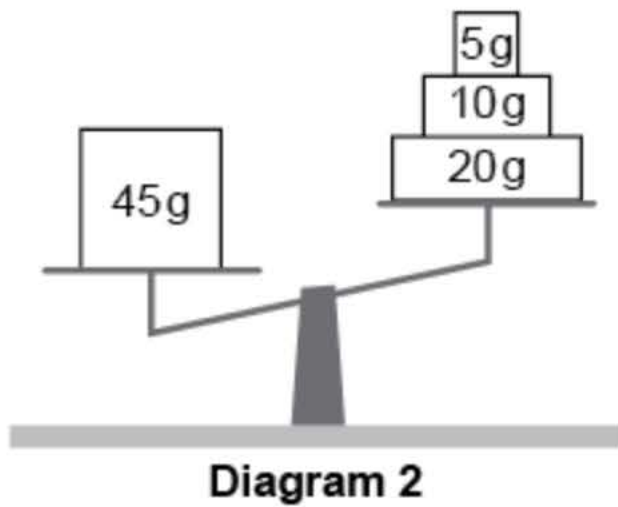
☐

Salt makes iron nails rust quickly.

Tilly balanced an object with some masses on a set of scales.



Tilly then replaced the object with another one that had a mass of 45 g.



What would Tilly have to do to balance the scales in diagram 2 so that both sides are level?

☐

Add 5 g to the left side of the scales in diagram 2.

☐

Add 10 g to the right side of the scales in diagram 2.

☐

Add 45 g to the right side of the scales in diagram 2.

☐

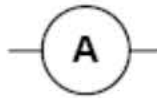
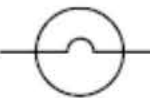
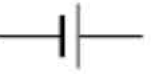
Add 10 g to the left side of the scales in diagram 2.

☐

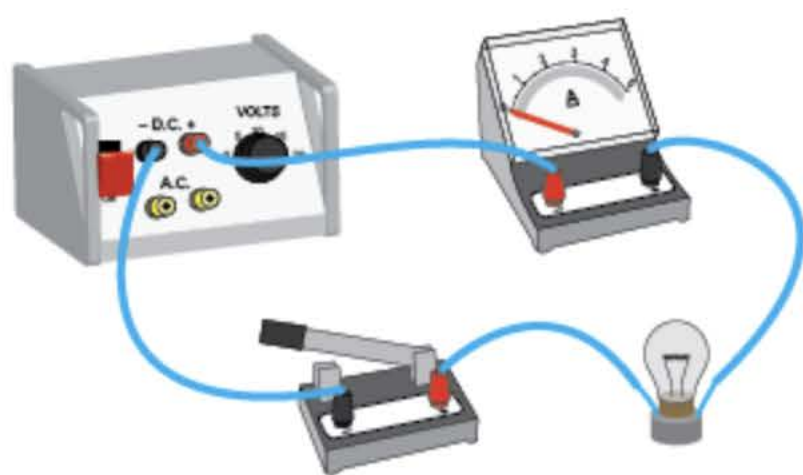
Add 20 g to the right side of the scales in diagram 2.

We use circuit symbols as an easy way to draw electrical circuits.

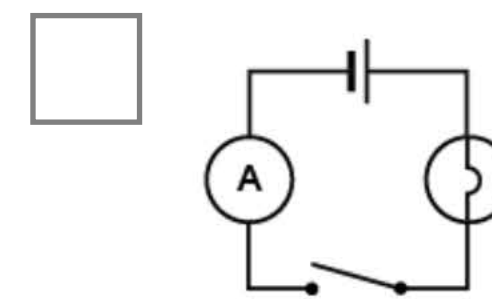
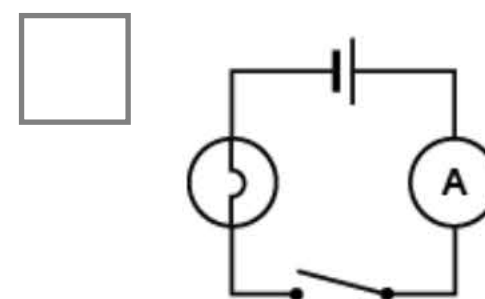
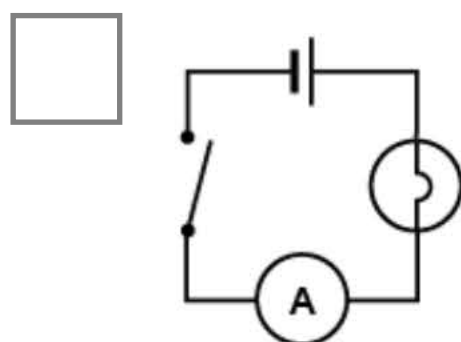
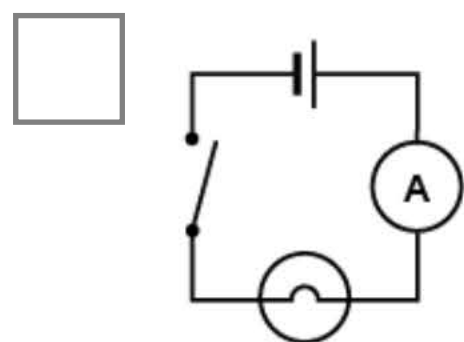
The table shows the symbols for some common electrical parts.

Part				
Symbol				

Amy made this circuit by joining some electrical parts together.



Which diagram represents Amy's circuit?



Animals that have a backbone are called vertebrates.

The table shows five groups of vertebrates and their characteristics.

	Fish	Amphibians	Reptiles	Birds	Mammals
Form of reproduction	lay eggs with no outer shell	lay eggs with no outer shell	lay eggs with outer shell	lay eggs with outer shell	give birth to live young
Body covering	interlocking scales	moist skin	interlocking scales	feathers	dry skin with hair
Method of breathing	through gills	through skin or gills; lungs as adult	through lungs	through lungs	through lungs
Cold- or warm-blooded	cold-blooded	cold-blooded	cold-blooded	warm-blooded	warm-blooded

Tuatara are animals that have interlocking scales and breathe through lungs.

Which statement about tuatara is true?

- ☐ They lay eggs with no outer shell.
- ☐ They are fish.
- ☐ They are reptiles.
- ☐ They are warm-blooded.

In the first photograph the arrow shows the length of a cicada's body.

In the second photograph the arrow shows the width of a cicada's thorax.



Body length



Thorax width

Which option has both the correct body length and correct thorax width?

- ☐ Body length is $4\frac{1}{2}$ cm Thorax width is $2\frac{1}{2}$ cm
- ☐ Body length is $4\frac{1}{2}$ cm Thorax width is 2 cm
- ☐ Body length is $6\frac{1}{2}$ cm Thorax width is $2\frac{1}{2}$ cm
- ☐ Body length is $6\frac{1}{2}$ cm Thorax width is 2 cm

Three students observed popcorn being made.



Kim observed: 'There were lots of little popping sounds and a smell like cooked butter.'

Peter observed: 'It was yellow in colour and had a round shape.'

Noor observed: 'It was hot in my hand and wasn't very heavy.'

Which statement about the observations is correct?

- ☐ Kim used her sense of smell and her sense of hearing.
- ☐ Peter used his sense of sight and his sense of taste.
- ☐ Kim and Noor both used their sense of smell.
- ☐ All three students used their sense of sight.

A key allows users to identify an object, such as an animal or a plant, using the characteristics of the object.

The diagram shows a key for a group of animals called vertebrates.

1.	has feathers	bird
	does not have feathers	go to 2
2.	has scales	go to 3
	does not have scales	go to 4
3.	has gills	fish
	does not have gills	reptile
4.	has fur	mammal
	does not have fur	amphibian

A caecilian does not have feathers, does not have scales and does not have fur.

What type of vertebrate is a caecilian?

☐

fish

☐

reptile

☐

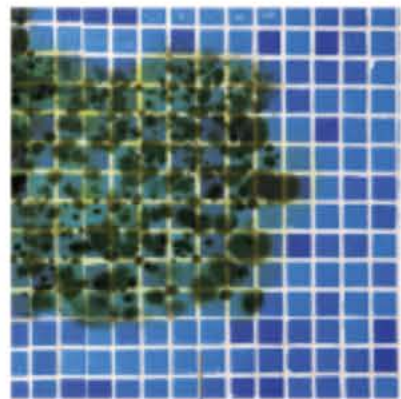
mammal

☐

amphibian

Angela wanted to test how well four different mould removal sprays worked. She sprayed each product on a mouldy surface in her home.

Product 1



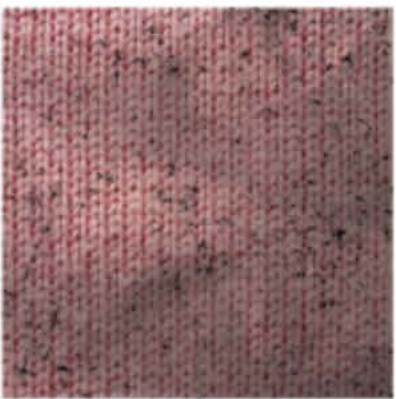
bathroom floor

Product 2



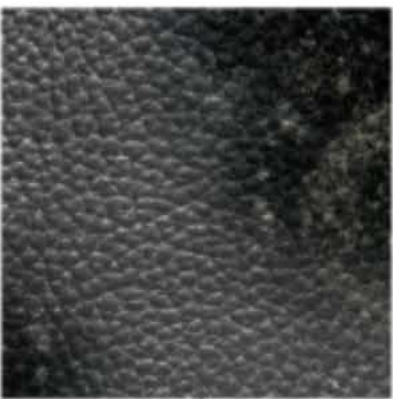
sink

Product 3



fabric

Product 4



leather

Angela left each spray in place for ten minutes before wiping the surfaces and comparing the results. Which of these options gives correct information about the investigation?

	Is this a fair test?	Why?
(A)	Yes	because different surfaces were tested
(B)	No	because 10 minutes was not long enough
(C)	Yes	because all methods to clean the surfaces were the same
(D)	No	because the products were not tested on the same surface

☐ A

☐ B

☐ C

☐ D

Soil erosion is the wearing away of the upper layer of soil.

Rainfall run-off is the rainwater that is not absorbed by the soil but flows across the soil surface.

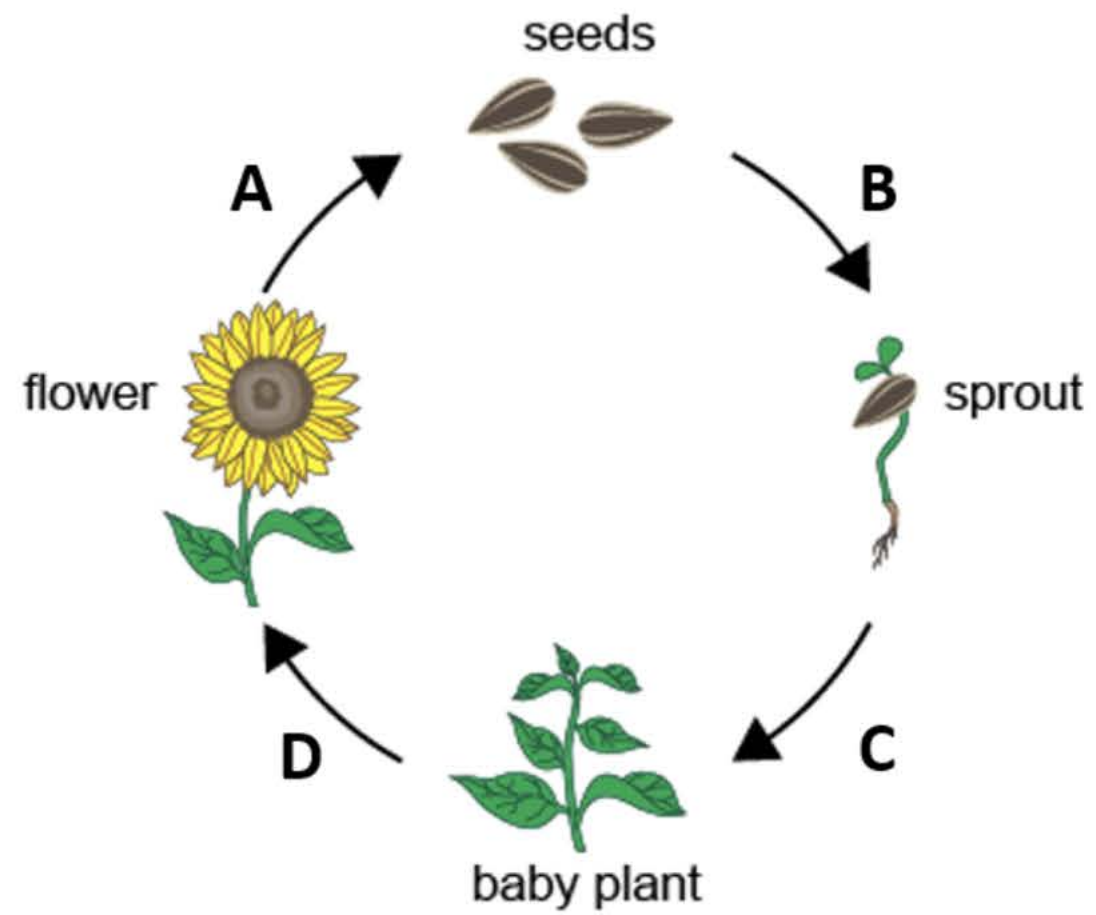
The table shows the results of an investigation into the effect of vegetation cover on soil erosion and rainfall run-off after a heavy storm.

Factor	Treatment area		
	A	B	C
Area of soil covered by vegetation (%)	87	69	6
Amount of rainfall run-off (mm)	3	26	70
Amount of soil lost (tonnes/hectare)	0.03	0.3	22

Which statement best summarises the results?

- ☐ High percentage of vegetation cover results in high amount of rainfall run-off.
- ☐ High amount of rainfall run-off results in low amount of soil lost.
- ☐ Low percentage of vegetation cover results in low amount of soil lost.
- ☐ Low percentage of vegetation cover results in high amount of soil lost.

Pollination is part of the process in which a plant produces seeds. The diagram shows the life cycle of a plant.



According to the diagram, at which stage does pollination take place?

☐

A

☐

B

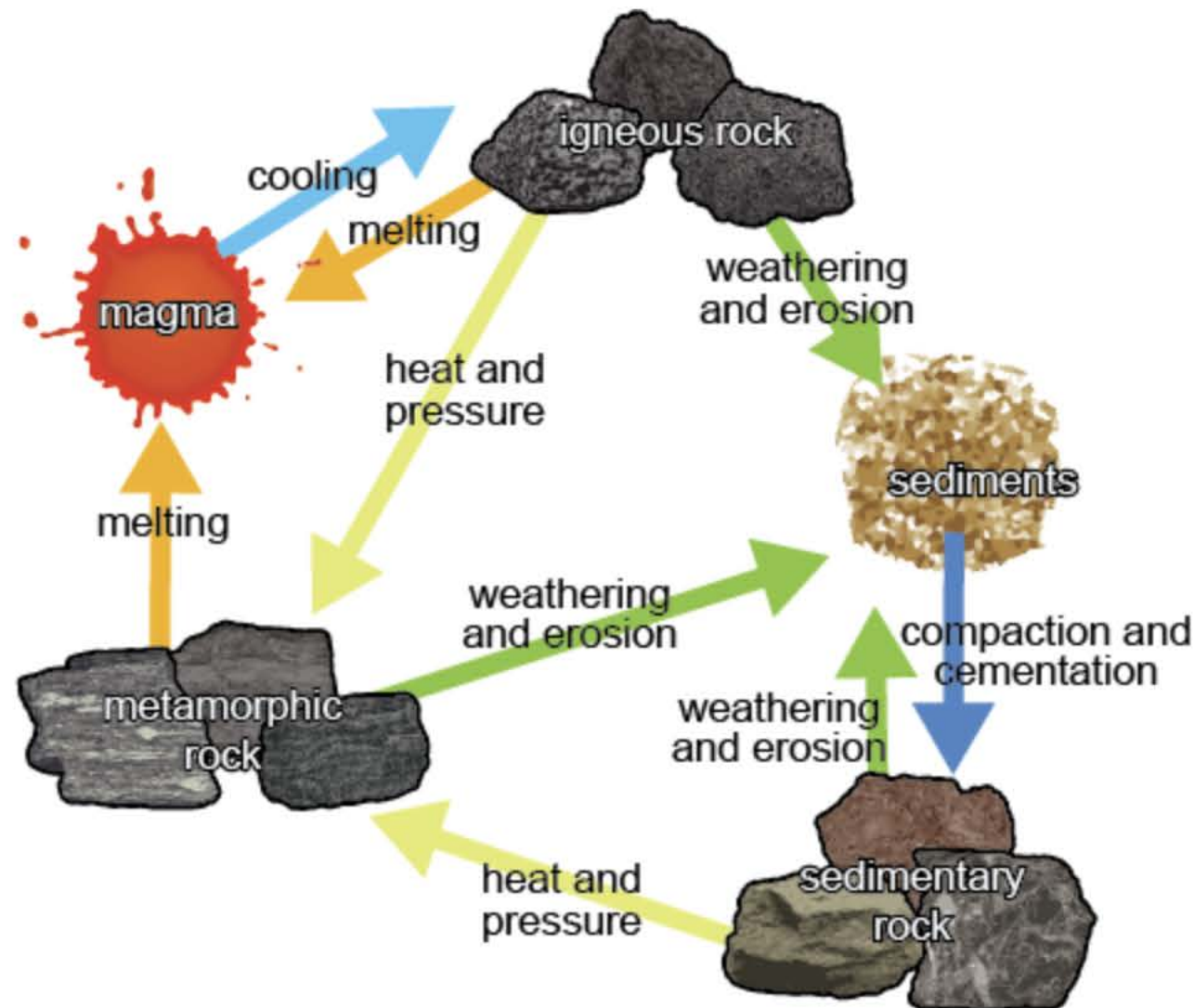
☐

C

☐

D

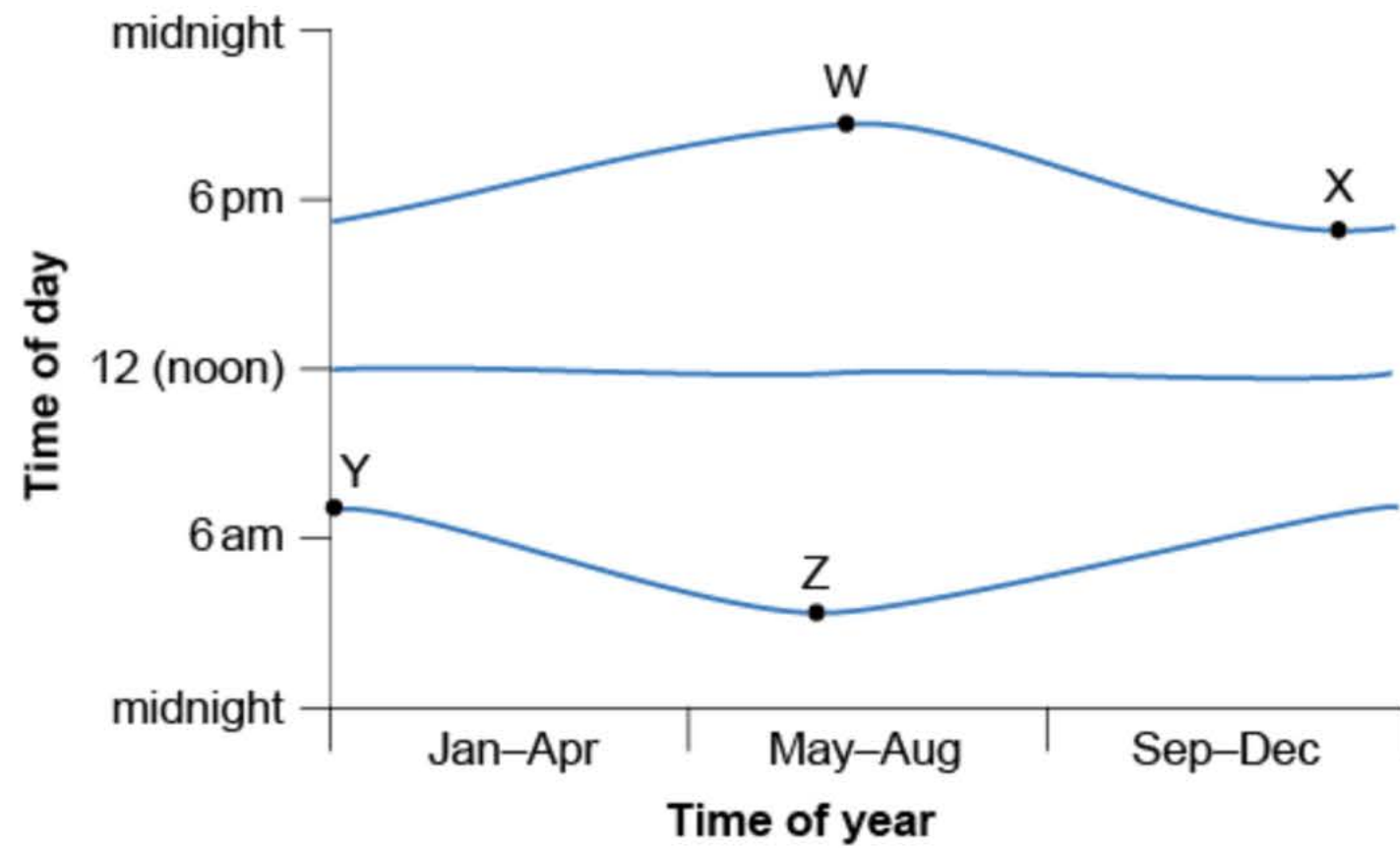
There are three main types of rocks: igneous, metamorphic and sedimentary rocks. The rock cycle diagram shows how geological processes convert rocks from one type to another type.



From the information in the diagram, which of these statements is **not** true?

- ☐ Magma can cool down to form any of the three types of rock.
- ☐ The formation of metamorphic rock requires the action of heat and pressure.
- ☐ Sediments can form when any of the three types of rock are eroded and weathered.
- ☐ The change of sediments into sedimentary rocks is a reversible process.

The graph shows the times of sunrise and sunset at one location on Earth over a year.



Which point represents the earliest sunset?

☐

W

☐

X

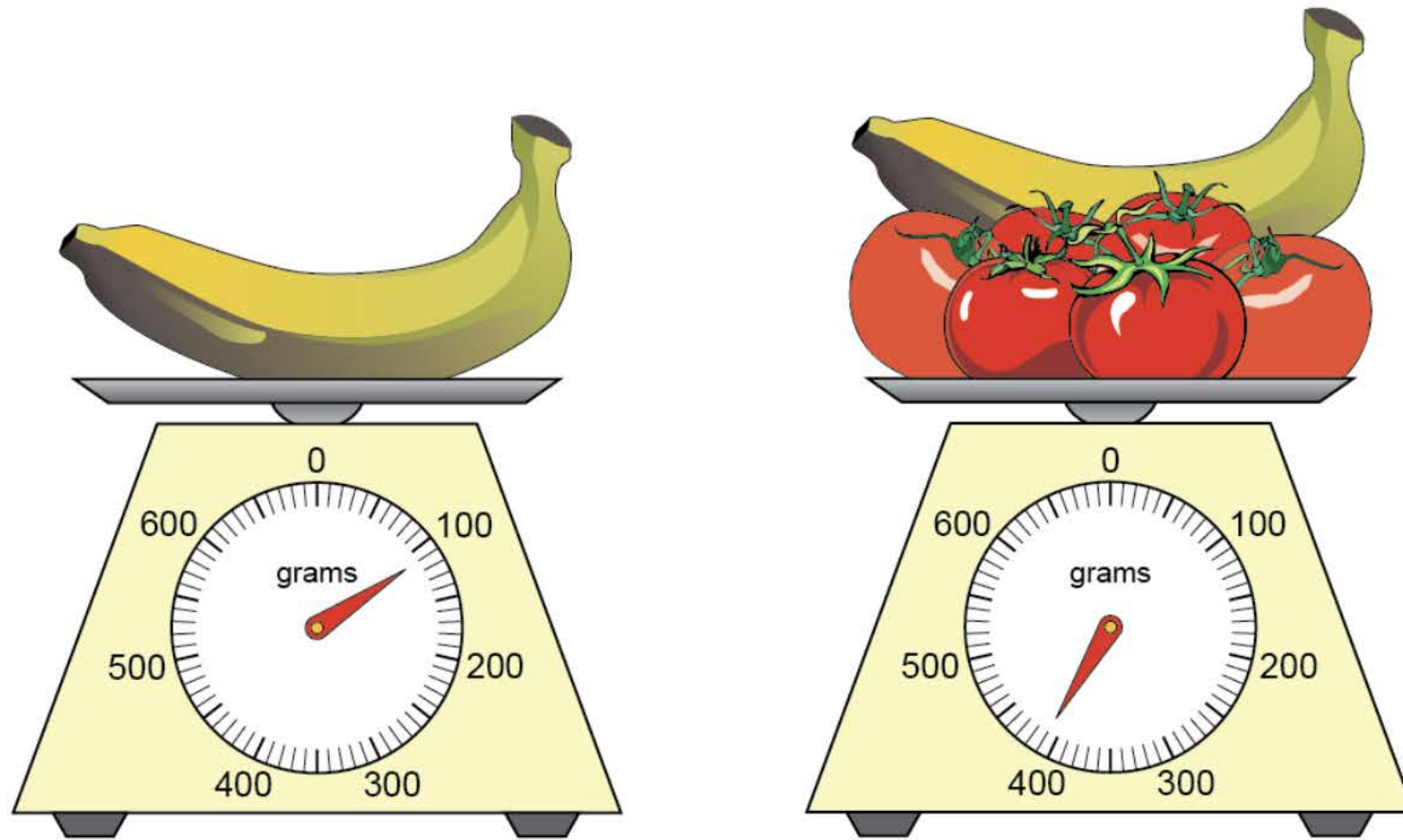
☐

Y

☐

Z

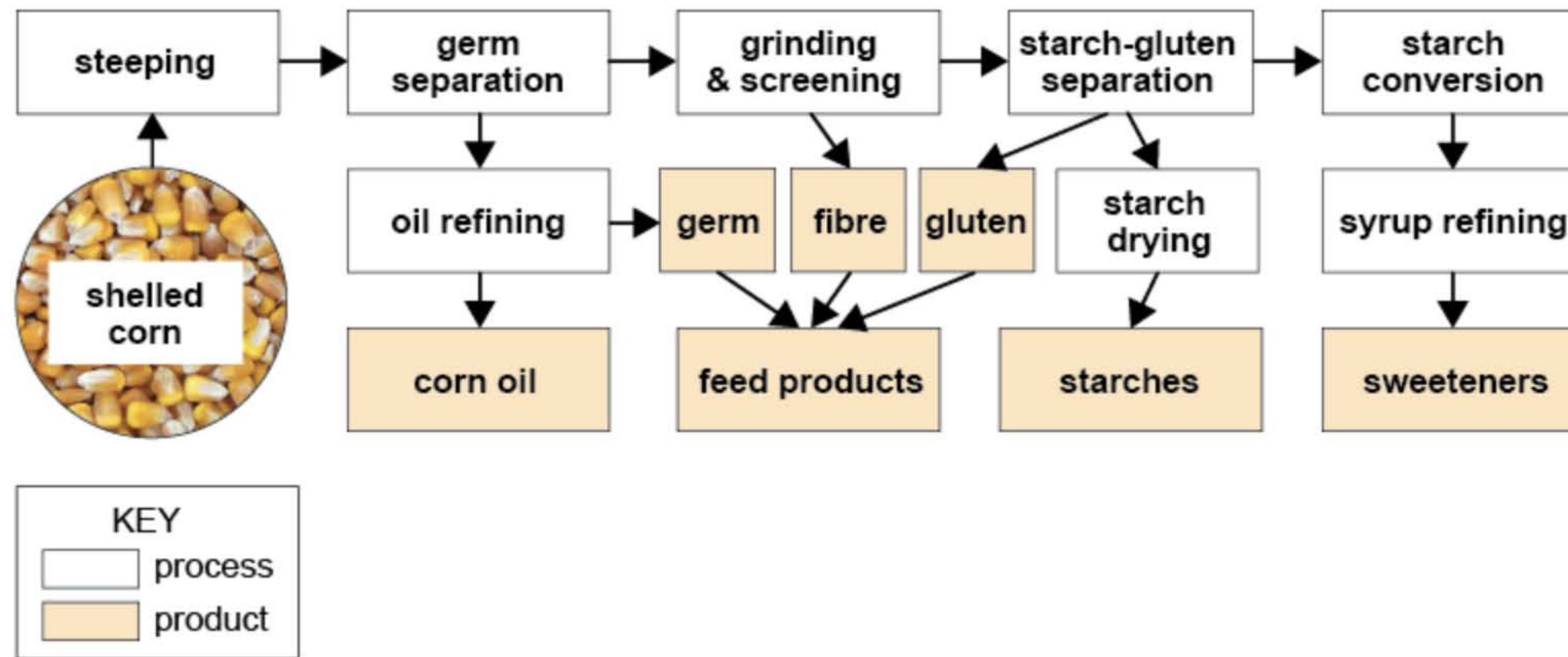
Sue weighed a banana and some tomatoes.



Approximately what is the mass of one tomato?

grams

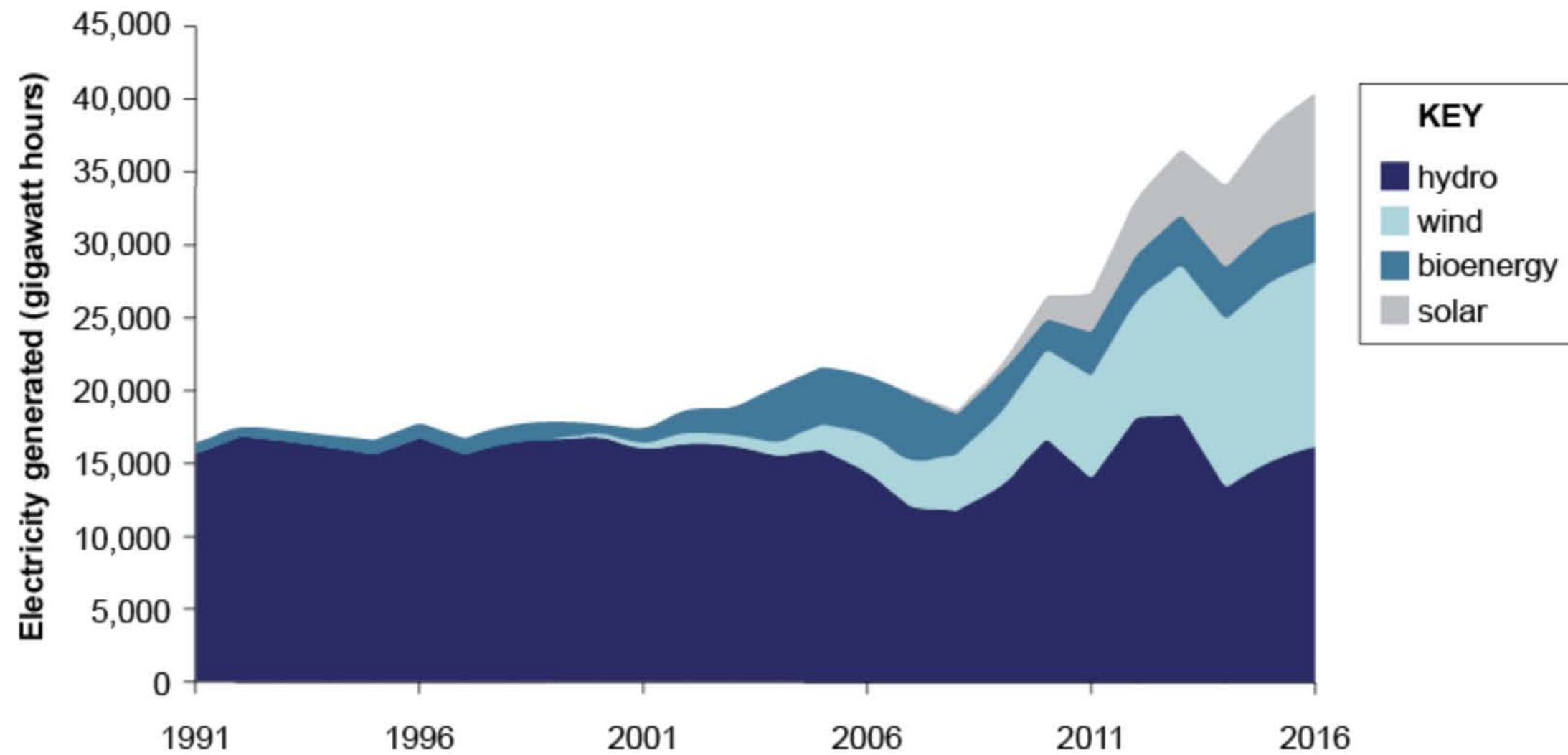
The diagram summarises the steps in producing a number of products from corn.



Which of these steps does **NOT** produce feed products?

- ☐ oil refining
- ☐ grinding and screening
- ☐ starch-gluten separation
- ☐ starch drying

The graph shows changes in the amounts of electrical energy generated from different renewable sources in Australia from 1991 to 2016.



Adapted electricity generation table, Australian Energy Update, Commonwealth of Australia 2018.

Which renewable source has seen its contribution increase the most from 2001 to 2016?

☐

hydro

☐

wind

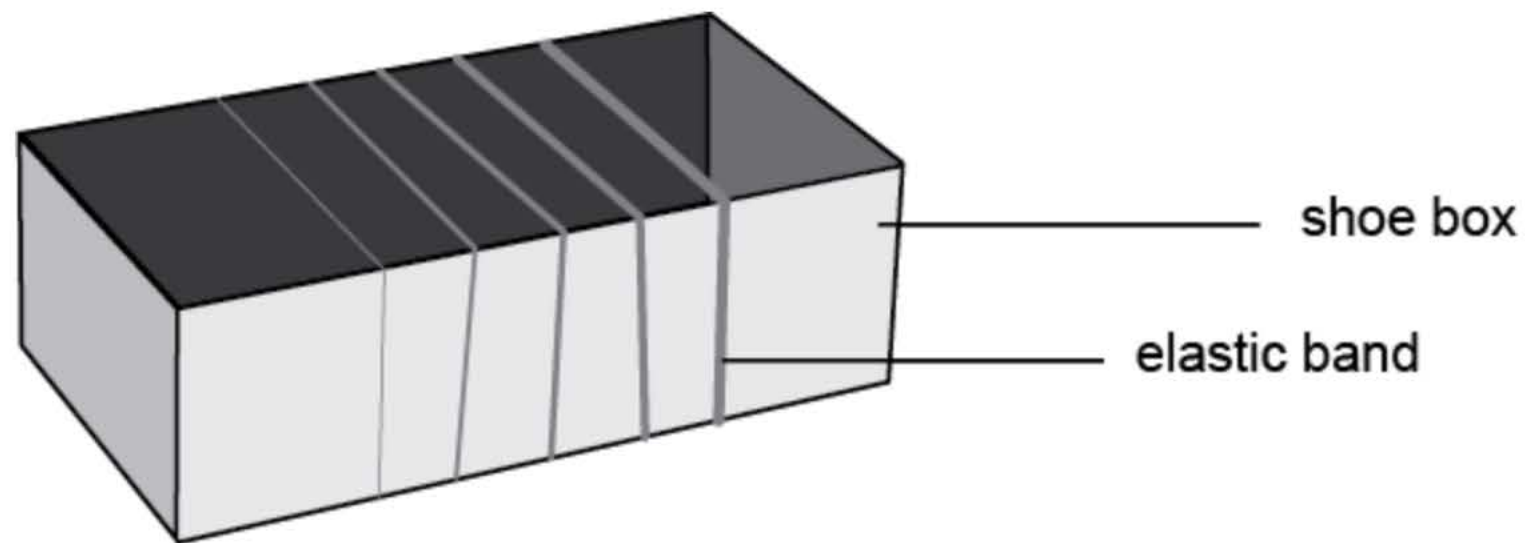
☐

bioenergy

☐

solar

Megan made a musical instrument using some elastic bands and an empty shoe box. The elastic bands all had different lengths and thicknesses. When she plucked them, each band made a sound with a different pitch.

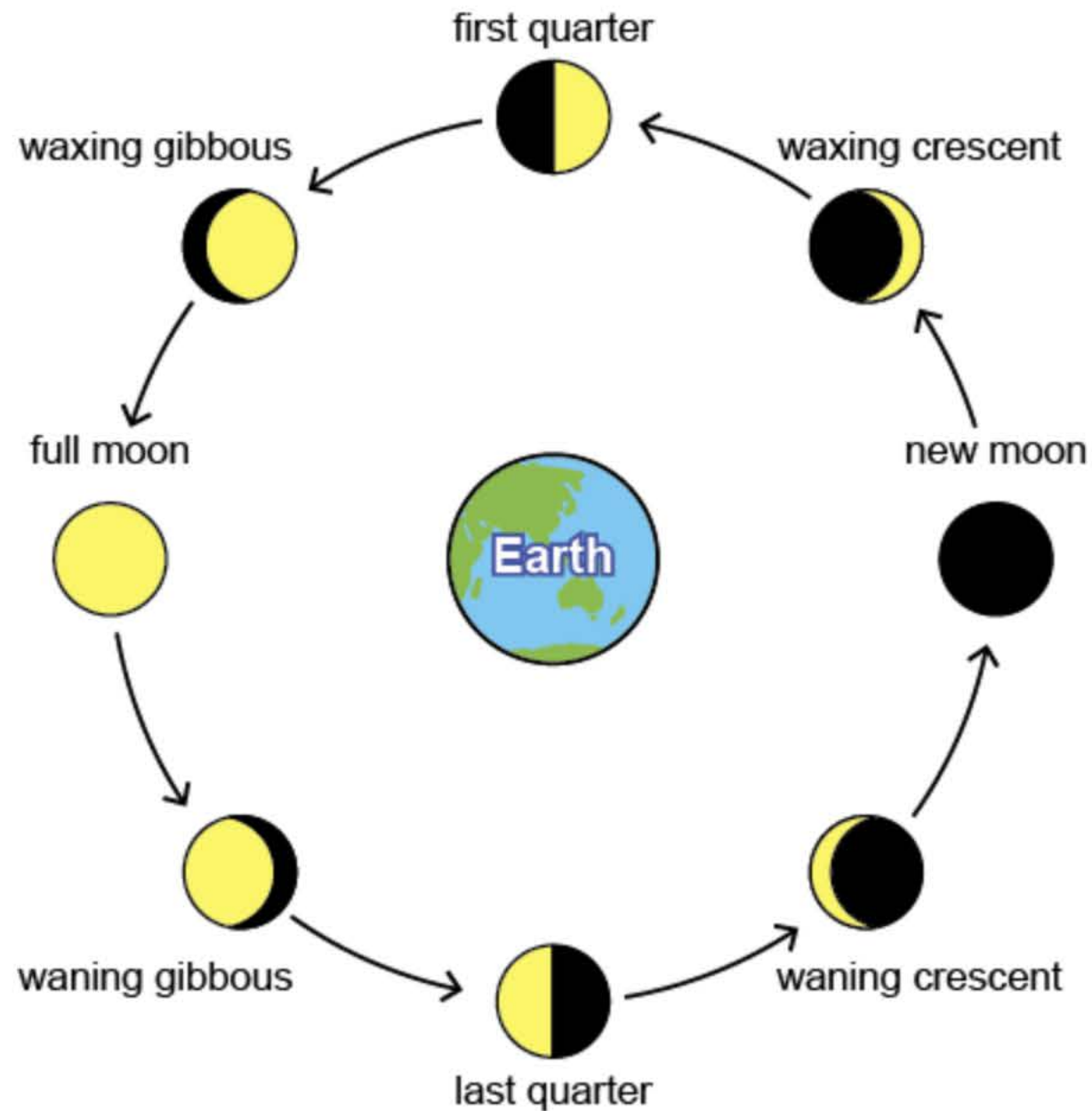


Megan was investigating whether the thickness of an elastic band affects the pitch of the sound it produces when plucked.

What must she do to make her investigation a fair test?

- ☐ pluck each elastic band three times
- ☐ use elastic bands made from different types of rubber
- ☐ use elastic bands with exactly the same thickness but different lengths
- ☐ use elastic bands with exactly the same length but different thicknesses

The phase of the Moon is the shape of the sunlit part of the Moon, as seen from Earth. The diagram shows the Moon's changing phases as the Moon orbits Earth.



Moon phase diagram by Andonee https://commons.wikimedia.org/wiki/File:Moon_Phase_Diagram_for_Simple_English_Wikipedia.GIF, CC BY-SA 3.0

The phases of the Moon

☐

get smaller from new moon to full moon.

☐

get bigger from first quarter to last quarter.

☐

get smaller then bigger from last quarter to first quarter.

☐

get smaller then bigger from full moon to new moon.

One serving of food is the amount of a type of food that an average person would normally eat in one meal. By measuring food in servings, it is possible to compare the nutrients and energy content (kJ) in foods.

The table outlines the energy content in one serving of some different foods.

Food	Serving size	Energy (kJ)
Soft drink	200 mL	360
Popcorn	1 cup	230
Meat pizza	1 slice	584
Fish burger	1 burger	820

According to the table, which of these has the greatest energy content?

☐

400 mL soft drink

☐

3 cups of popcorn

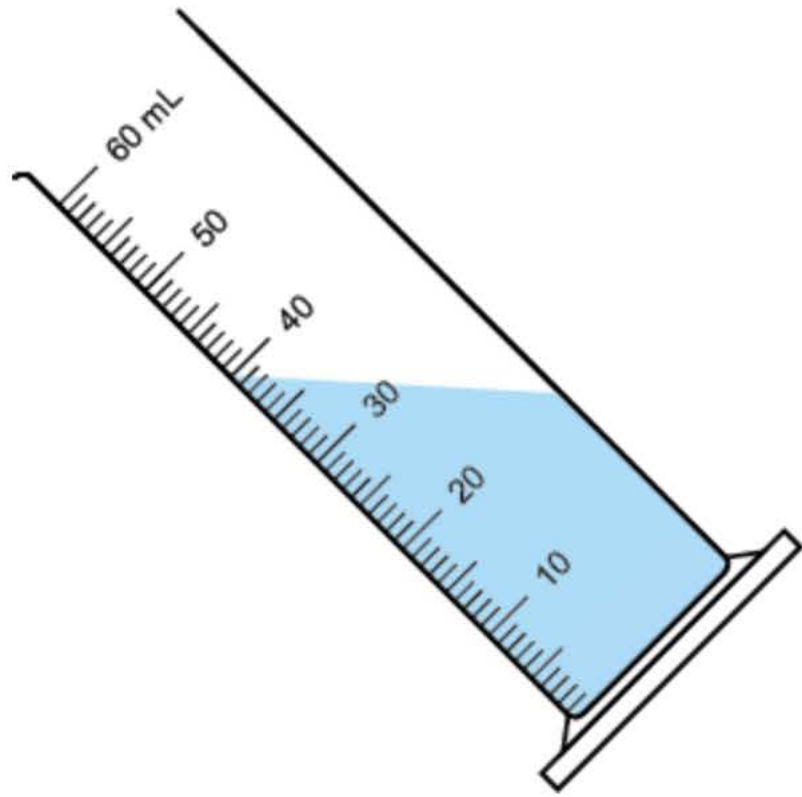
☐

1 fish burger

☐

2 slices of meat pizza

Tara poured some water into a measuring cylinder and then tilted the measuring cylinder as shown.



How much water (mL) is in the measuring cylinder?

☐

20

☐

30

☐

35

☐

40

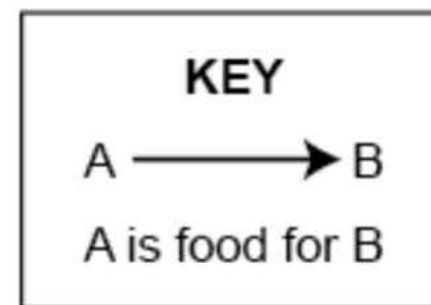
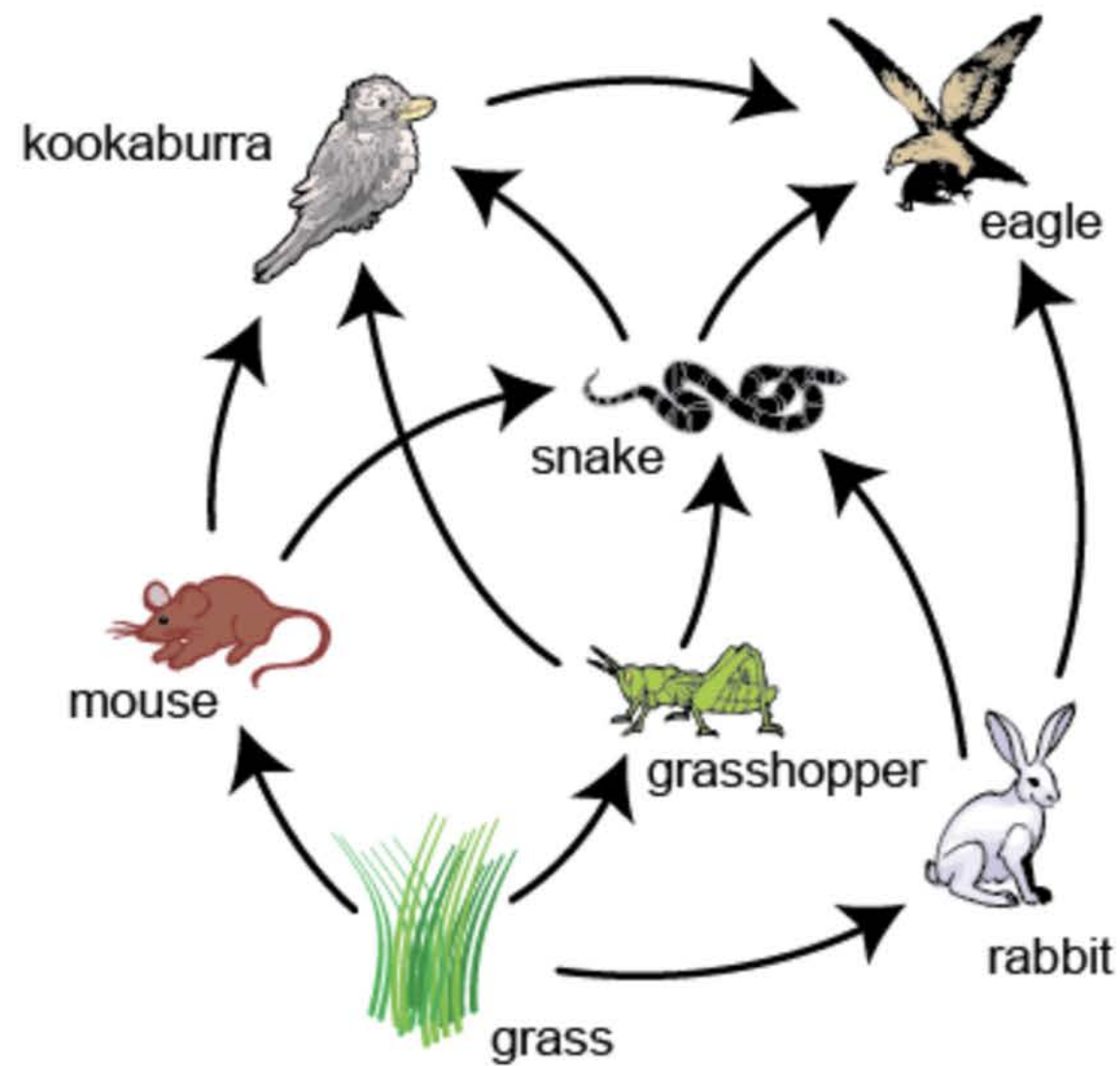
The table classifies flowers by their heights when fully grown.

Short		Medium		Tall	
Name	Height (cm)	Name	Height (cm)	Name	Height (cm)
Grecian windflower	5 – 15	Lily of the valley	15 – 30	Spanish bluebell	25 – 46
Siberian squill	12 – 20	Jonquilla daffodil	25 – 40	Lily-flowered tulip	36 – 60
Dwarf iris	10 – 15	Hyacinth	20 – 30	Short cup daffodil	30 – 46

Which statement is correct based on the table?

- ☐ A flower with a height of 34 cm when fully grown must be a tall flower.
- ☐ A flower with a height of 15 cm when fully grown must be a short flower.
- ☐ A flower with a height of 20 cm when fully grown must be a short flower.
- ☐ A flower with a height of 23 cm when fully grown must be a medium flower.

This is a food web that shows how the energy in food moves through a group of living things in a particular area.



Living things that eat plants and animals are called consumers.

- Primary consumers feed on plants.
- Secondary consumers feed on primary consumers.
- Tertiary consumers feed on primary consumers and secondary consumers.

Which of the following animals is a secondary consumer in the food web?

☐

eagle

☐

kookaburra

☐

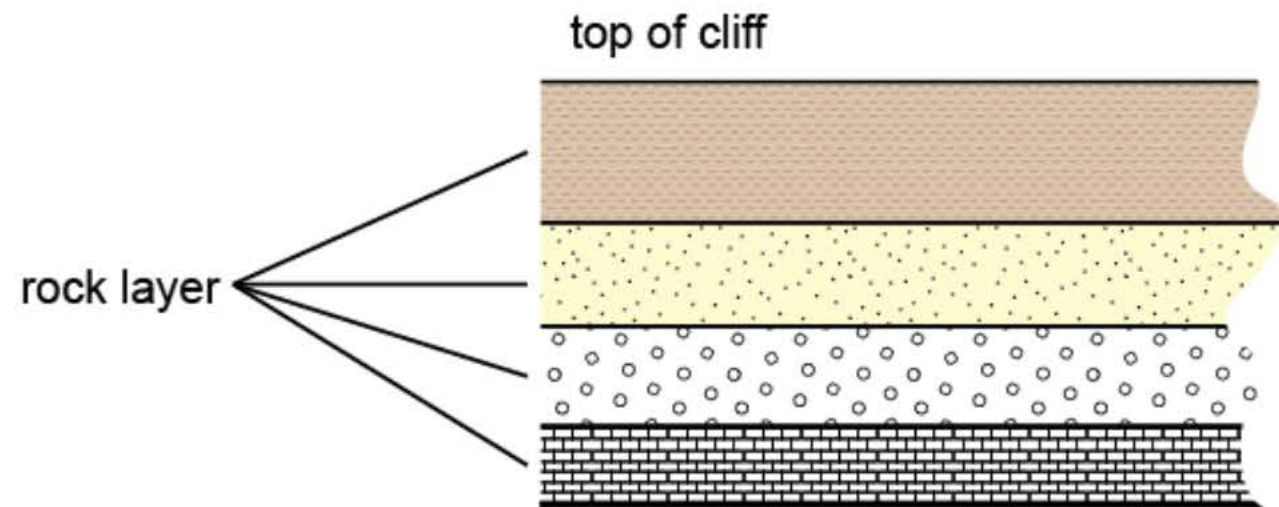
rabbit

☐

snake

The law of superposition states that layers of rock in an undisturbed sequence of rocks are arranged in a time sequence, with the oldest layer at the bottom and the newest layer on top.

The diagram shows part of a cliff face with four different layers of rock.



We know the following information about the layers:

1. Shale is younger than conglomerate.
2. Limestone is the oldest.

Which option could be the correct key for the cliff face?

☐

KEY	
	shale
	sandstone
	conglomerate
	limestone

☐

KEY	
	shale
	limestone
	sandstone
	conglomerate

☐

KEY	
	limestone
	conglomerate
	shale
	sandstone

☐

KEY	
	conglomerate
	shale
	sandstone
	limestone

Nitinol is a special material made by mixing two metals.

Four students conducted an investigation using a paperclip made of Nitinol.

original
paperclip-shaped
Nitinol in water at
room temperature



water at room
temperature

students fold
the Nitinol



no water

students add hot
water to the
random-shaped
Nitinol



hot water

Nitinol changes
back to paperclip
shape

Images of paperclip-shaped Nitinol by Petermaerki https://commons.wikimedia.org/wiki/File:Nitinol_bueroklammer_verbogen.jpg and https://commons.wikimedia.org/wiki/File:Nitinol_bueroklammer_heiss.jpg, CC BY-SA 3.0.

When they had finished the investigation, the students made the following remarks:

- Eloise said: 'I think if we heat the Nitinol more, it will melt.'
- Shaun said: 'The paperclip was easy to bend at room temperature.'
- Peter said: 'Nitinol wire is used in dental braces to push the teeth to the desired arrangement.'
- Rauli said: 'The folded random-shaped piece of Nitinol reverted to its original paperclip shape when we heated it.'

Whose statement is a prediction?

☐

Eloise

☐

Shaun

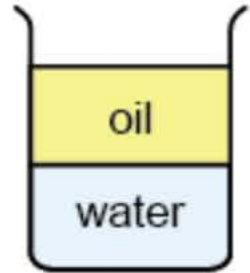
☐

Peter

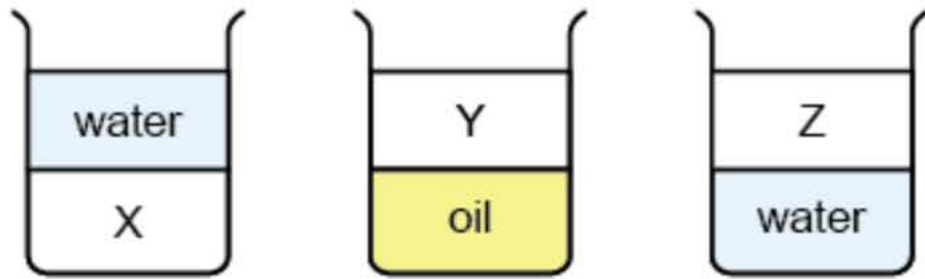
☐

Rauli

When two liquids do not mix, they separate into layers with the less dense liquid on top of the more dense one. For example, water and oil do not mix and oil is less dense than water.



Three different liquids X, Y and Z do not mix with water or oil. The diagram shows what happens when each liquid is added to water or oil.



Which statement is supported by this information?

☐

Z is less dense than Y.

☐

X is more dense than Y and Z.

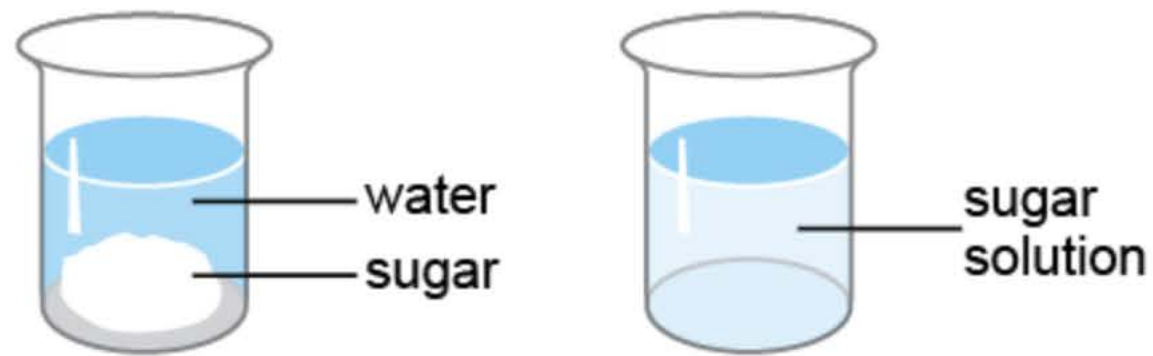
☐

X and Z are as dense as water.

☐

Y and Z are more dense than oil.

When you add some sugar to a glass of water and stir for a while, the sugar crystals slowly disappear, leaving a clear liquid. We say the sugar dissolves in water, and the two together make a sugar solution.



The table shows how much sugar can dissolve in 100 grams of water at different temperatures.

Temperature (°C)	10	20	40	60
Amount of sugar (grams)	180	200	240	280

How many grams of sugar can dissolve in 50 grams of water at 20 °C?

☐

100

☐

150

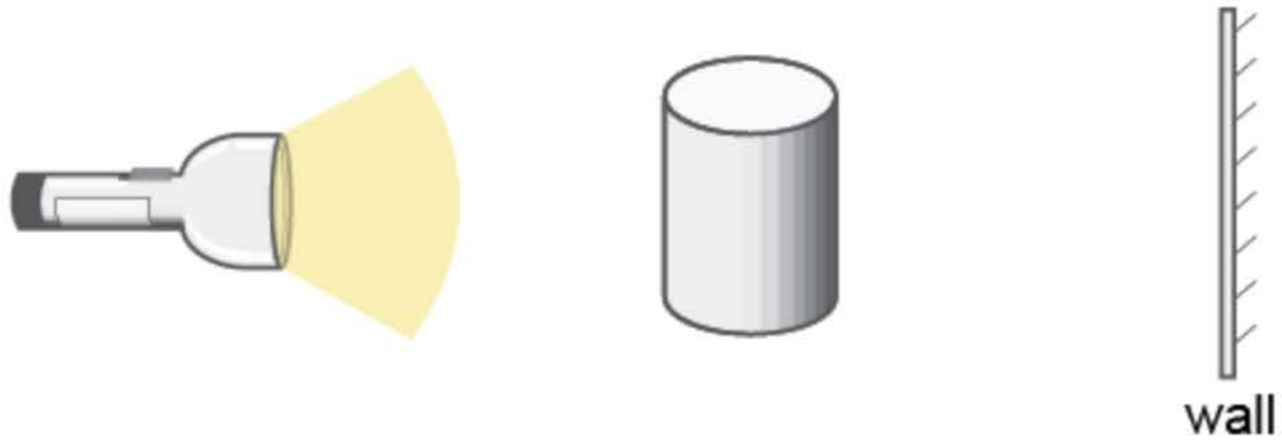
☐

200

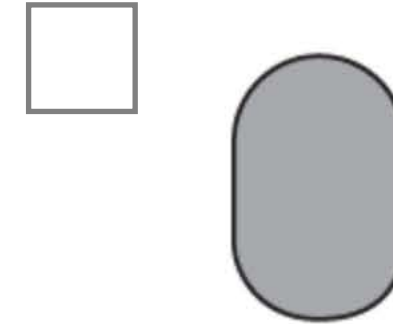
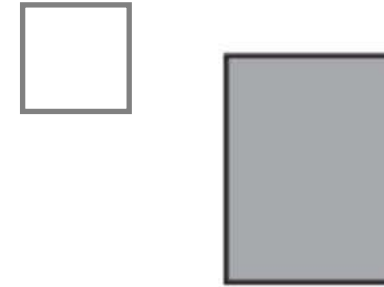
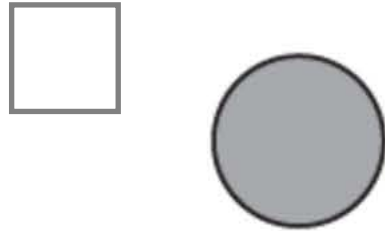
☐

250

Joyce shines a torch onto a cylinder, as shown.

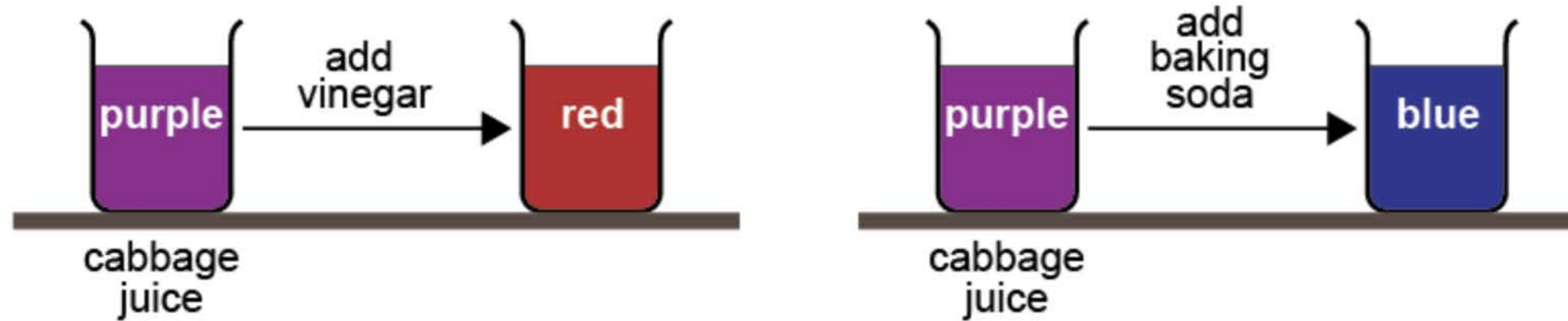


Which diagram shows the shadow produced by the cylinder on the wall?



Purple cabbage juice can be used to indicate whether a substance is alkaline or acidic.

Max took two cups of purple cabbage juice. He then added vinegar to one and baking soda to the other and noted the colour changes.

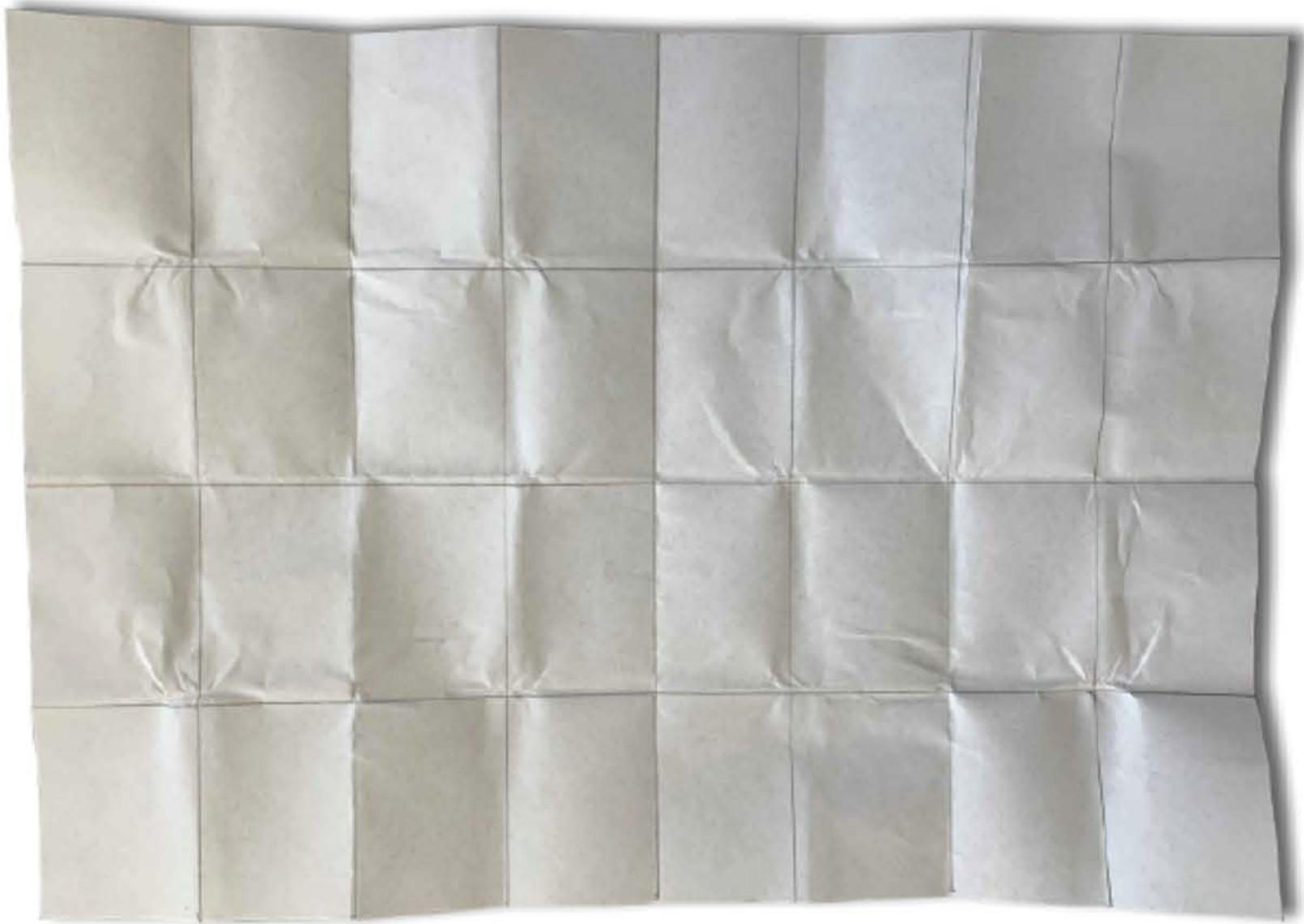


Max knew that vinegar is acidic and baking soda is alkaline. Max added some liquid he found in the kitchen to a new cup of cabbage juice. The cabbage juice turned red.

What can Max conclude from his experiment?

- ☐ The cabbage juice was acidic.
- ☐ The liquid contained vinegar.
- ☐ The liquid was acidic.
- ☐ The liquid was red.

Adam folded an unused piece of paper in half several times. The photograph shows what his paper looked like when he unfolded it.



How many times in total had he folded the paper?

☐

4

☐

5

☐

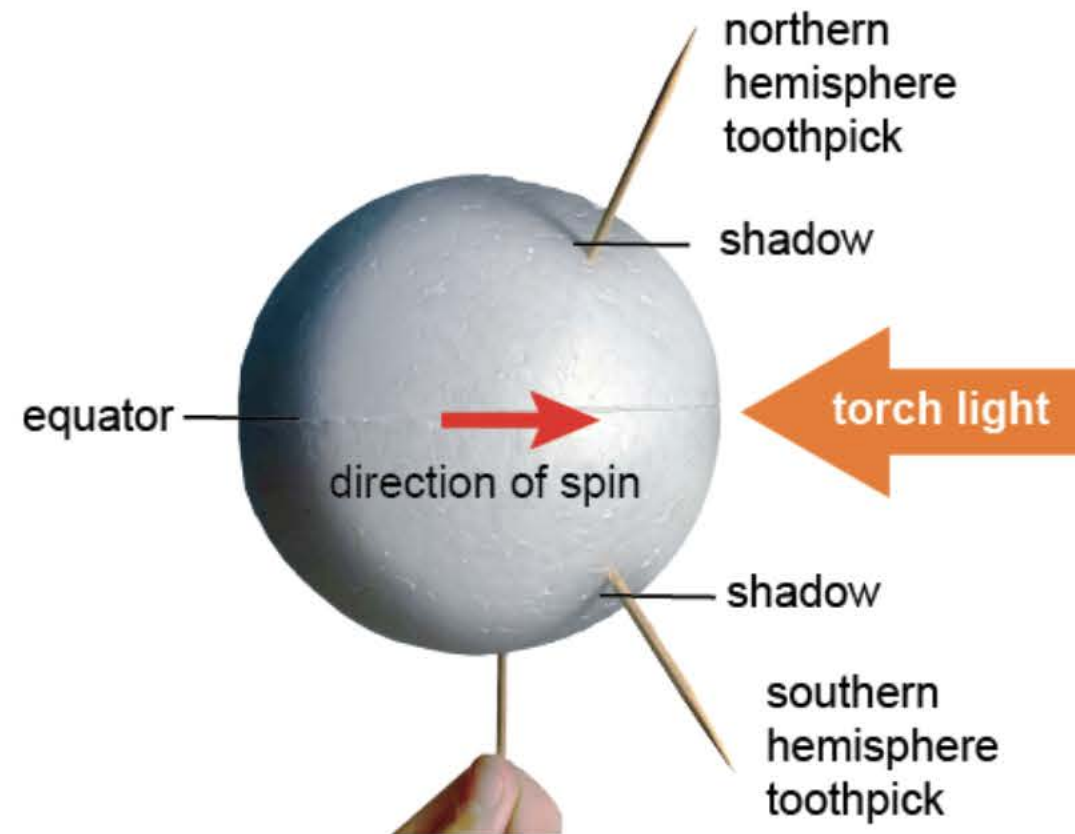
8

☐

16

Millie and Tom investigated shadows using a styrofoam ball on a stick to represent Earth.

They pushed one toothpick into the northern hemisphere and one toothpick into the southern hemisphere.



In a dark room, Tom shone a torch at the equator.

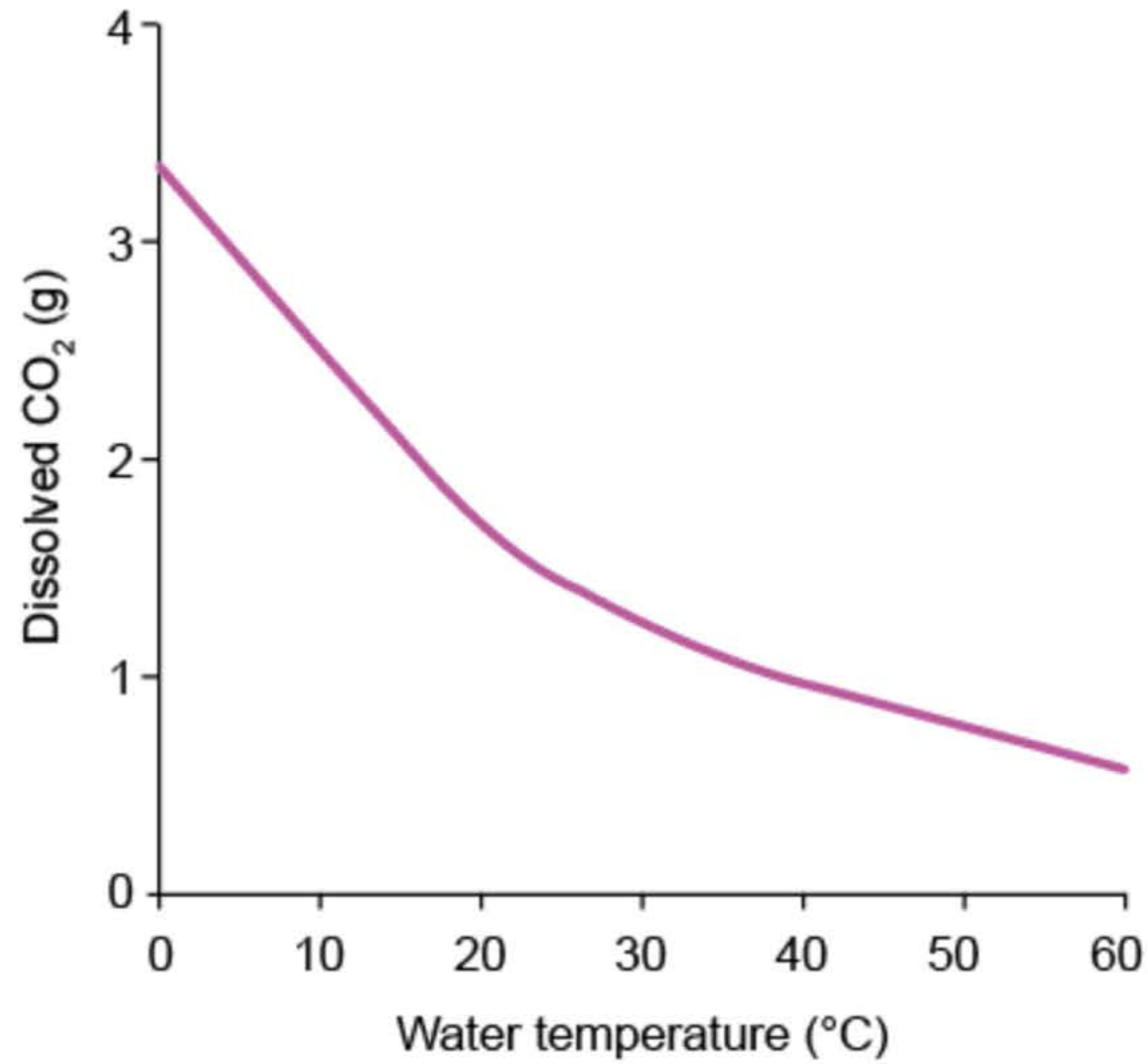
Millie twisted the stick to make the Earth spin in the direction shown.

The toothpick shadows moved either clockwise  or anticlockwise  around each toothpick.

Which of the following statements correctly describes the movement of the toothpicks' shadows?

- ☐ The northern hemisphere toothpick shadow moves clockwise and the southern hemisphere toothpick shadow moves clockwise.
- ☐ The northern hemisphere toothpick shadow moves clockwise and the southern hemisphere toothpick shadow moves anticlockwise.
- ☐ The northern hemisphere toothpick shadow moves anticlockwise and the southern hemisphere toothpick shadow moves anticlockwise.
- ☐ The northern hemisphere toothpick shadow moves anticlockwise and the southern hemisphere toothpick shadow moves clockwise.

The graph shows the amount of carbon dioxide gas (CO_2) that dissolves in 1 kg of water at different temperatures.



Which statement is supported by the data in the graph?

- ☐ The higher the water temperature, the less CO_2 dissolves.
- ☐ Water temperature has no effect on the amount of CO_2 that dissolves.
- ☐ CO_2 only dissolves when the water temperature is below 60 °C.
- ☐ The lower the water temperature, the less CO_2 dissolves.

A SCOBY is a symbiotic culture of bacteria and yeast. Kombucha is made by adding a SCOBY to sugar and black tea and leaving the mixture to ferment. The SCOBY is alive and grows in a thin layer across the top of the tea.

Lian wanted to find out if increasing the amount of sugar would increase the thickness of the SCOBY.

She set up four new jars similar to the one in the photo. Each new jar contained:

- a SCOBY that was 2 mm thick
- 200 mL of tea brewed from 3 tea bags of black tea
- 30 g sugar.

Then she added 5 g more sugar to Jar 2, 10 g more sugar to Jar 3, and 15 g more sugar to Jar 4.

Lian placed all four jars on the kitchen bench. After two weeks, Lian measured the thickness of the SCOBY in each jar.

Jar	Thickness of SCOBY (mm)	
	Start	2 weeks
1	2	5
2	2	5
3	2	5
4	2	5

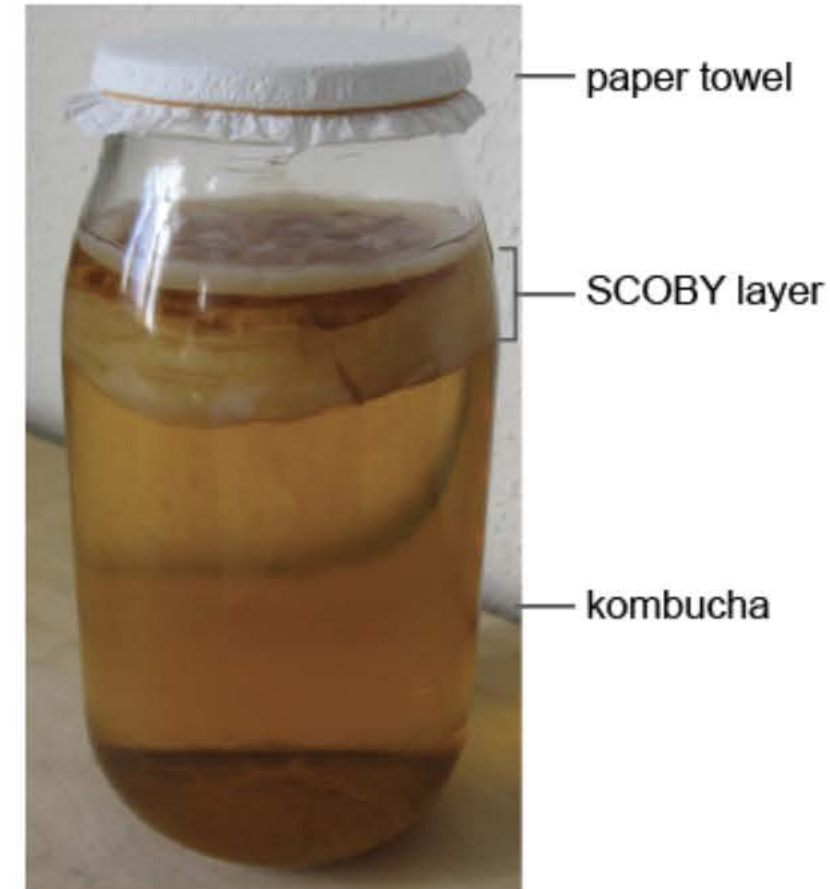
What can Lian conclude from her results?

☐ The thickness of the SCOBY will stay the same whether sugar is added to the jar or not.

☐ Adding more than 45 g of sugar will increase the thickness of the SCOBY.

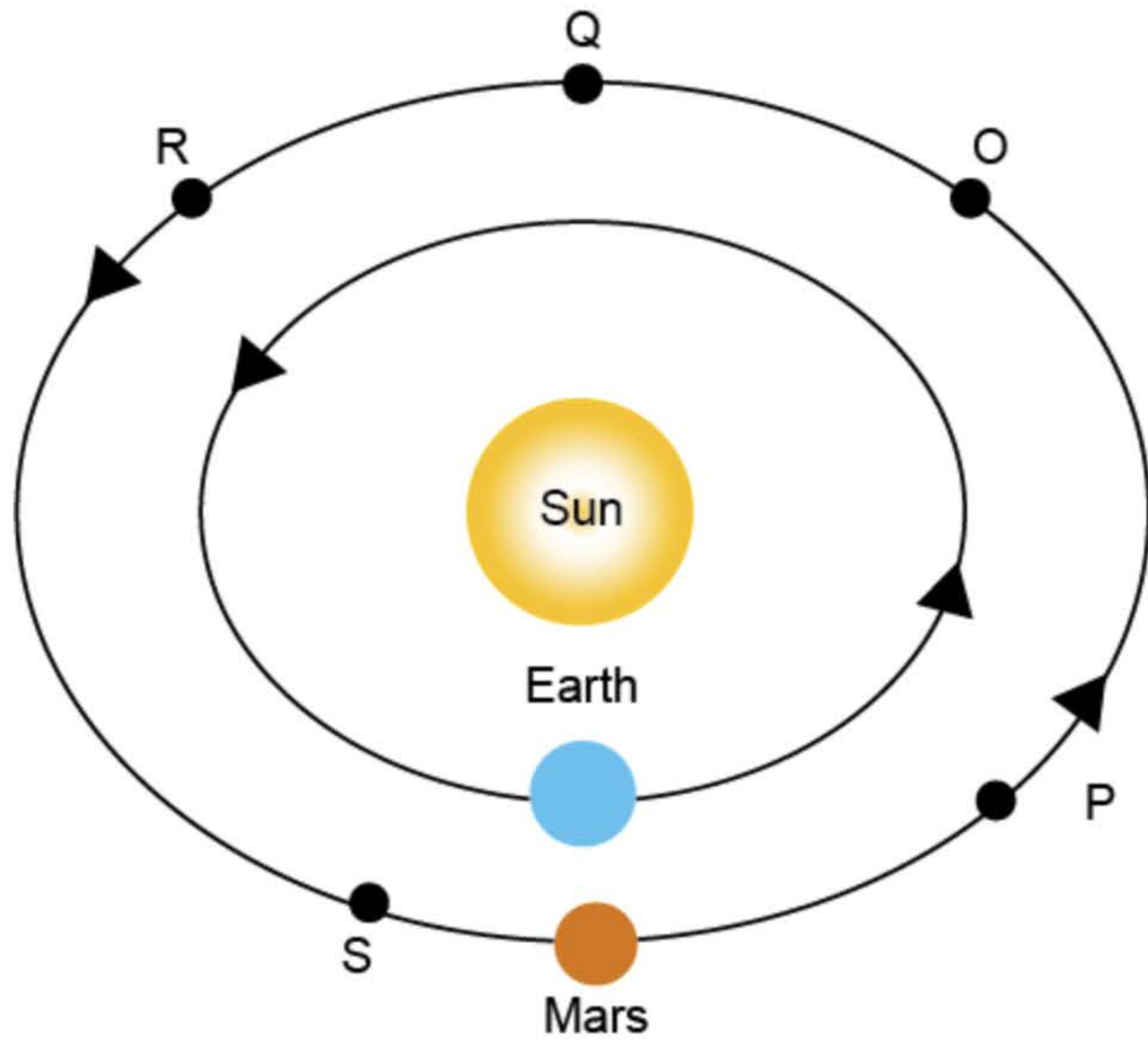
☐ The more sugar that is added to the jar, the thicker the SCOBY becomes.

☐ The thickness of the SCOBY will stay the same when 30–45 g of sugar is added.



Kombucha image by Bjoern Appel <https://commons.wikimedia.org/wiki/File:Kombucha.jpg>, CC BY-SA 3.0.

It takes Mars approximately two Earth years to orbit the Sun once.



Where was Mars eight months ago?

☐

P

☐

Q

☐

R

☐

S

☐

O

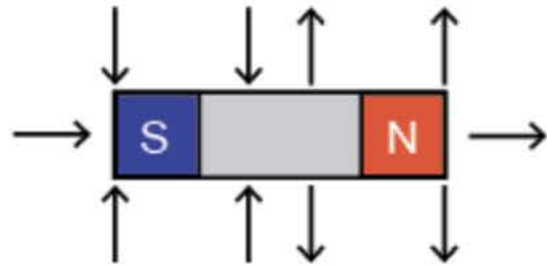
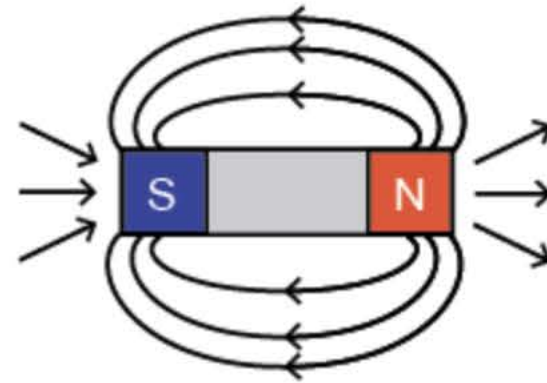
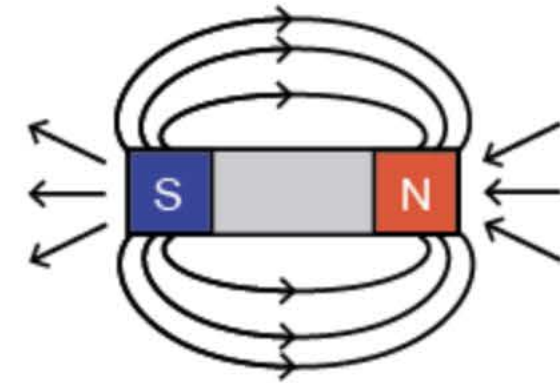
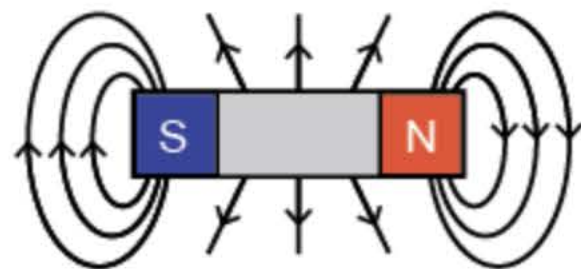
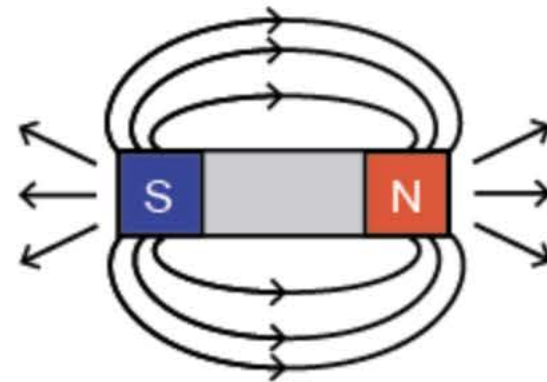
A magnet has two opposite poles, north and south.

The photo shows the magnetic field, marked by iron filings, of a bar magnet.



The direction of the field is outward from the north pole and in towards the south pole of the magnet.

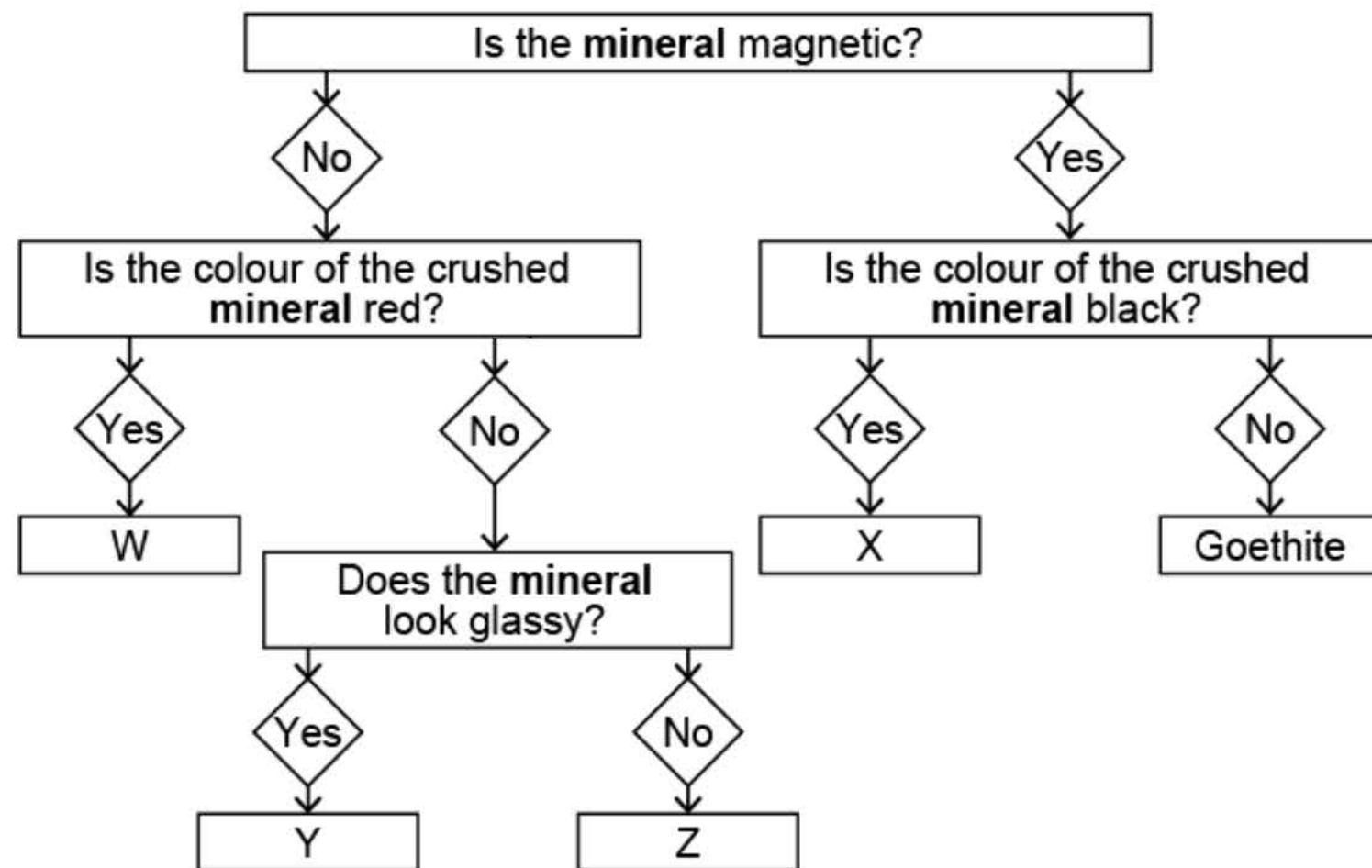
Which diagram correctly shows the magnetic field of the bar magnet?

☐

☐

☐

☐

☐


Minerals are natural substances found in rocks. The table below shows some properties of minerals containing iron.

Mineral	Colour of crushed mineral	Magnetic?	Appearance	Iron content (%)
Haematite	red	no	metallic	70
Goethite	brown/orange	slightly	dull	63
Limonite	yellow/brown	no	earthy	63
Magnetite	black	strongly	metallic	74
Siderite	white	no	glassy	48

Using the table, Pedro set up this key to help identify some minerals.



Which of the following correctly identifies four of the minerals?

☐

W: siderite
X: limonite
Y: magnetite
Z: haematite

☐

W: haematite
X: magnetite
Y: siderite
Z: limonite

☐

W: haematite
X: limonite
Y: magnetite
Z: siderite

☐

W: limonite
X: magnetite
Y: siderite
Z: haematite

Results

Question number	Correct answer	Mark	Category	Descriptor
1	D	☐ ☐	Predicting & Concluding	Identify what affects the amount of energy captured by solar panels
2	C	☐ ☐	Observing & Measuring	Estimate half of a length given a full length
3	D	☐ ☐	Interpreting	Interpret a column graph about a student allergy survey
4	C	☐ ☐	Investigating	Identify the conditions for a fair test involving parachutes
5	B	☐ ☐	Predicting & Concluding	Predict the movement of the water level in response to an action
6	C	☐ ☐	Observing & Measuring	Compare skull diagrams from different views
7	A	☐ ☐	Investigating	Deduce what can be known from an investigation
8	B	☐ ☐	Reasoning & Problem solving	Determine how to balance a weighing scale
9	A	☐ ☐	Interpreting	Match a circuit diagram with a built circuit
10	C	☐ ☐	Predicting & Concluding	Draw a conclusion from information in a table
11	B	☐ ☐	Observing & Measuring	Measure lengths using a ruler
12	A	☐ ☐	Observing & Measuring	Identify senses used in an observation
13	D	☐ ☐	Predicting & Concluding	Use a key to classify a caecilian
14	D	☐ ☐	Investigating	Identify whether a described investigation was a fair test
15	D	☐ ☐	Interpreting	Interpret results of an investigation into soil erosion
16	A	☐ ☐	Interpreting	Interpret a diagram of a plant life cycle
17	A	☐ ☐	Predicting & Concluding	Draw a conclusion based on a diagram of the rock cycle
18	B	☐ ☐	Interpreting	Interpret an unfamiliar type of graph
19	50	☐ ☐	Observing & Measuring	Determine the average mass of a number of objects
20	D	☐ ☐	Interpreting	Identify the relationship between elements of a flowchart
21	B	☐ ☐	Predicting & Concluding	Draw a conclusion based on information from a graph
22	D	☐ ☐	Investigating	Identify factors that make a test fair
23	C	☐ ☐	Predicting & Concluding	Conclude how the Moon phases change shape using information in a diagram
24	D	☐ ☐	Reasoning & Problem solving	Draw a conclusion based on information in a table
25	B	☐ ☐	Observing & Measuring	Estimate the volume of water in a measuring cylinder
26	D	☐ ☐	Predicting & Concluding	Identify correct statement based on information in a table about the heights of plants
27	D	☐ ☐	Interpreting	Interpret a food web to categorise consumers
28	A	☐ ☐	Reasoning & Problem solving	Determine the order of rock layers in a cliff face
29	A	☐ ☐	Investigating	Identify the statement that is a prediction
30	B	☐ ☐	Reasoning & Problem solving	Determine relationships by logical comparison

Results

Question number	Correct answer	Mark	Category	Descriptor
31	A		Interpreting	Perform a calculation using data from a table
32	C		Reasoning & Problem solving	Predict the shape of the shadow cast by a given object
33	C		Investigating	Draw a conclusion based on the results of an investigation
34	B		Reasoning & Problem solving	Determine the number of times a piece of paper was folded
35	B		Reasoning & Problem solving	Solve a shadow problem by spatial reasoning
36	A		Interpreting	Identify the trend in a line graph
37	D		Predicting & Concluding	Conclude from data on energy source of a SCOBY
38	C		Reasoning & Problem solving	Solve a problem based on planetary orbits
39	B		Reasoning & Problem solving	Determine the correct diagram of a magnetic field
40	B		Interpreting	Interpret tabled information to construct a dichotomous key

You have completed this practice test.

Your mark is

0 / 40

0%

Click here to reset the test and try again.

